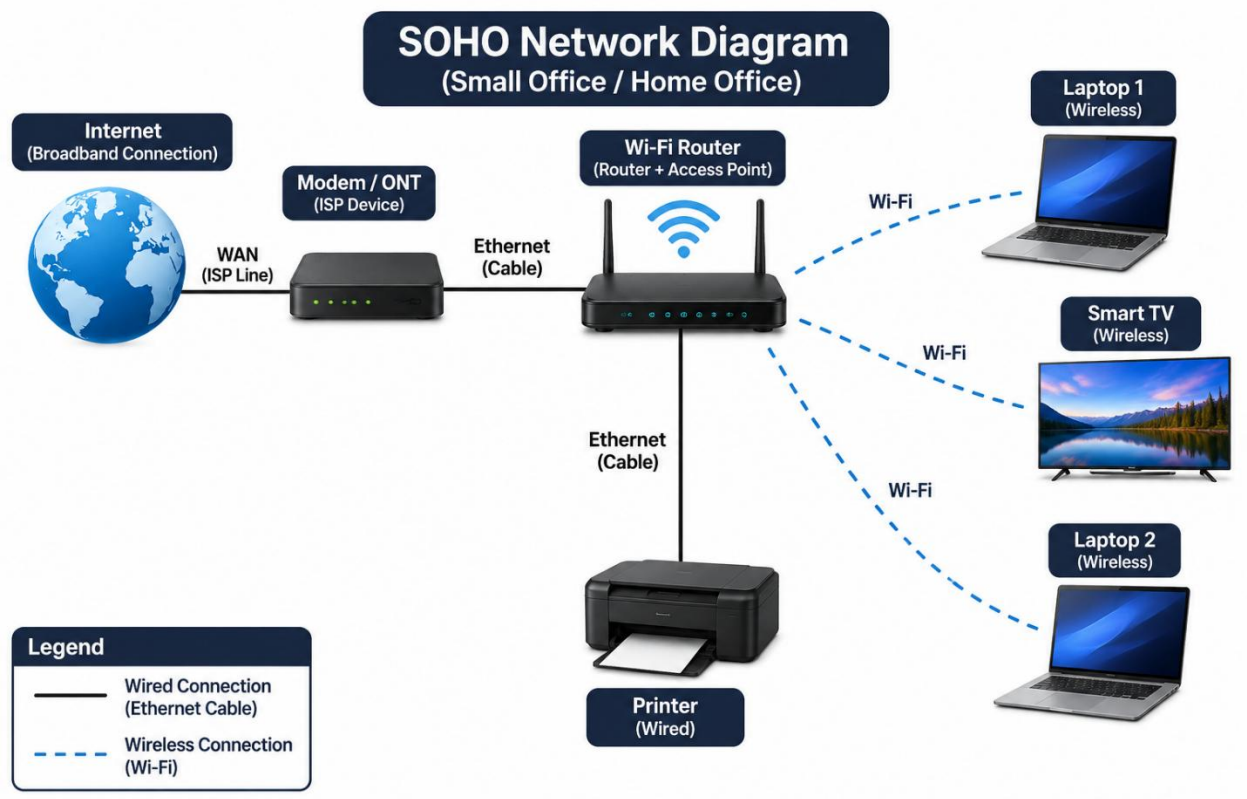
Build a SOHO network

Task 1 : Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).



Device	Connection Type
Internet to Modem	Broadband
Modem to Router	Wired (Ethernet)
Laptop 1 to Router	Wireless (Wi-Fi)
Laptop 2 to Router	Wireless (Wi-Fi)
Smart TV to Router	Wireless (Wi-Fi)
Printer to Router	Wired (Ethernet)

Description

The broadband connection provides internet access to the SOHO network through a modem. The modem is connected to a Wi-Fi router using an Ethernet cable. Two laptops, one Smart TV and one printer connect to the router wirelessly using Wi-Fi.

The Wi-Fi router allows all connected devices to communicate and access the internet.

Task 2 : List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.

To identify all the hardware components required to physically build the proposed SOHO network, including suitable brand and model suggestions.

1. Hardware Components List

No.	Hardware Component	Suggested Brand / Model	Quantity
1	Broadband Modem / ONT	Airtel / JioFiber ONT	1
2	Wi-Fi Router	TP-Link Archer AX23	1
3	Laptop 1	HP / Dell / Lenovo	1
4	Laptop 2	HP / Dell / Lenovo	1
5	Smart TV	Samsung / LG	1
6	Wireless Printer	HP Smart Tank 529	1
7	Ethernet Cable	Cat6 RJ45 Cable	1
8	Power Adapters	Supplied with devices	As required
9	UPS (Optional)	APC / Microtek	1

2. Description of Hardware Components

1. Broadband Modem / ONT:
It receives the internet connection from the Internet Service Provider (ISP), such as Airtel or JioFiber.

2. Wi-Fi Router – TP-Link Archer AX23:
It distributes the internet connection to multiple devices through wireless Wi-Fi connectivity.

3. Laptops – HP / Dell / Lenovo:
Two laptops are used to access the internet and perform office or personal work.

4. Smart TV – Samsung / LG:
The Smart TV connects to the Wi-Fi network to access online streaming services.

5. Wireless Printer – HP Smart Tank 529:
It connects to the Wi-Fi network and allows users to print documents wirelessly.

6. Ethernet Cable – Cat6 RJ45:
It provides a wired connection between the broadband modem/ONT and the Wi-Fi router.

7. Power Adapters:
They provide electrical power to the network devices.

8. UPS – APC / Microtek (Optional):
It provides temporary backup power during electricity failures.

3. Additional Network Devices

Device	Required?	Reason
Network Switch	No	The router supports wireless connections for all four devices.
Wi-Fi Extender	No	Not required if the router provides sufficient Wi-Fi coverage.
UPS	Optional	Provides backup power during electricity failures.

Conclusion

The above hardware components are sufficient to build the proposed SOHO network. A broadband ONT, one Wi-Fi router, two laptops, one Smart TV and one wireless printer are required. A Cat6 Ethernet cable connects the ONT to the router, while the other devices connect wirelessly through Wi-Fi.

A network switch and Wi-Fi extender are not required for this basic SOHO setup.

Task 3 : Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

To configure the Wi-Fi router for the SOHO network by setting a unique SSID, enabling WPA2 security, changing the default administrator password and enabling DHCP.

Step-by-Step Router Configuration

Step 1: Connect the Router

1. Connect the broadband modem/ONT to the WAN port of the Wi-Fi router using an Ethernet cable.
2. Connect the router to the power supply.
3. Switch on the router and wait for it to start.

Step 2: Access the Router Administration Page

1. Connect a laptop to the router using Wi-Fi or an Ethernet cable.
2. Open a web browser.
3. Enter the router's IP address, such as 192.168.0.1, in the address bar.
4. Enter the login credentials provided by the router manufacturer.
5. Open the router's administration panel.

Step 3: Change the Default Administrator Password

1. Open the Administration or System Settings section.
2. Select the Change Password option.
3. Enter the existing administrator password.
4. Create a new strong administrator password.
5. Save the settings.

This helps protect the router from unauthorized access.

Step 4: Set a Unique SSID (Wi-Fi Network Name)

1. Open Wireless Settings.
2. Find the SSID or Network Name option.

3. Enter a unique network name.

Example SSID: SOHO_Office_Network

4. Save the settings.

The SSID helps users identify their Wi-Fi network.

Step 5: Enable WPA2 Security

1. Open Wireless Security Settings.
2. Select WPA2-Personal (WPA2-PSK).
3. Select AES encryption.
4. Set a strong Wi-Fi password.

Example Wi-Fi Password: Office@Net2026!

5. Save the settings.

WPA2 security protects the wireless network from unauthorized access.

Step 6: Enable DHCP

1. Open the DHCP Server settings.
2. Enable the DHCP Server option.
3. Set the IP address range:
    - Starting IP Address: 192.168.1.100
    - Ending IP Address: 192.168.1.199
4. Set the default gateway to 192.168.1.1.
5. Save the settings.

DHCP automatically assigns IP addresses to connected devices.

Step 7: Connect All Devices to Wi-Fi

1. Open Wi-Fi settings on both laptops.
2. Select SOHO_Office_Network.
3. Enter the Wi-Fi password.
4. Connect the Smart TV and wireless printer to the same Wi-Fi network.

5. Check whether all devices can access the network and internet.

Router Configuration Summary

Setting	**Configuration**
Router Model	TP-Link Archer AX23
Router IP Address	192.168.1.1
SSID	SOHO_Office_Network
Security Type	WPA2-Personal
Encryption	AES
Wi-Fi Password	Office@Net2026!
DHCP Server	Enabled
DHCP Range	192.168.1.100–192.168.1.199

Conclusion

The Wi-Fi router is successfully configured by setting a unique SSID, enabling WPA2 security, changing the default administrator password and enabling DHCP. All network devices can now connect to the SOHO network and access the internet securely.

Task 4 : Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

To configure IP addresses for all devices in the SOHO network by assigning static IP addresses to the printer and Smart TV and using DHCP for both laptops.

1. Network Configuration Details

The SOHO network uses a private IPv4 address range of 192.168.1.0/24.

Network Setting	Value
Network Address	192.168.1.0
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	8.8.8.8
DHCP Range	192.168.1.100 – 192.168.1.199

2. IP Address Assignment Table

The following table shows the MAC addresses and IP addresses assigned to each device. The MAC addresses are fictional examples created for this assignment.

Device	MAC Address	Assigned IP Address	Method
Wi-Fi Router	00:1A:2B:3C:4D:01	192.168.1.1	Static
Printer	00:1A:2B:3C:4D:02	192.168.1.10	Static
Smart TV	00:1A:2B:3C:4D:03	192.168.1.20	Static
Laptop 1	00:1A:2B:3C:4D:04	192.168.1.100	DHCP
Laptop 2	00:1A:2B:3C:4D:05	192.168.1.101	DHCP

3. Static IP Configuration

Static IP addresses are manually assigned to devices so that they keep the same IP address.

A. Printer Configuration

1. Open the printer's Network Settings.

2. Select Manual IP Configuration.
3. Enter IP Address: 192.168.1.10.
4. Enter Subnet Mask: 255.255.255.0.
5. Enter Default Gateway: 192.168.1.1.
6. Save the settings.

B. Smart TV Configuration

1. Open the Smart TV's Network Settings.
2. Select IP Settings.
3. Change the IP setting from Automatic to Manual.
4. Enter IP Address: 192.168.1.20.
5. Enter Subnet Mask: 255.255.255.0.
6. Enter Default Gateway: 192.168.1.1.
7. Save the settings.

4. DHCP Configuration for Laptops

DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses to network devices.

Steps:

1. Open Network Settings on the laptop.
2. Select Wi-Fi Properties.
3. Open IPv4 Settings.
4. Select Obtain an IP Address Automatically.
5. Select Obtain DNS Server Address Automatically.
6. Save the settings.

The Wi-Fi router automatically assigns IP addresses to both laptops from the DHCP range.

Example:

- Laptop 1: 192.168.1.100

- Laptop 2: 192.168.1.101

5. Verification of IP Addresses

To check the assigned IP address on a Windows laptop:

1. Open Command Prompt.
2. Type the following command:

ipconfig

3. Press Enter.
4. Check the IPv4 Address, Subnet Mask and Default Gateway.

To check network connectivity, use:

ping 192.168.1.1

If replies are received, the laptop can communicate with the router.

Conclusion

The SOHO network is configured using static IP addresses for the printer and Smart TV, while both laptops receive IP addresses automatically through DHCP. This configuration makes network management easier and helps avoid IP address conflicts.

Task 5 : Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

To identify and troubleshoot possible internet connectivity problems in a SOHO network.

If one of the laptops cannot connect to the internet, I would check the following possible causes and apply the appropriate solutions.

1. Incorrect Wi-Fi Password

Problem: The laptop cannot connect to the Wi-Fi network because the password is incorrect.

Solution:

1. Open Wi-Fi Settings on the laptop.
2. Forget the saved Wi-Fi network.
3. Connect again using the correct Wi-Fi password.
4. Check whether the internet is working.

2. Weak Wi-Fi Signal

Problem: The laptop is too far from the router, or walls and other obstacles are affecting the Wi-Fi signal.

Solution:

1. Move the laptop closer to the Wi-Fi router.
2. Check the Wi-Fi signal strength.
3. Restart the router if necessary.
4. Use a Wi-Fi extender if the coverage is insufficient.

3. Incorrect IP Address or DHCP Problem

Problem: The laptop does not receive a valid IP address from the router.

Solution:

1. Open Command Prompt.
2. Type the following command:

ipconfig /release

3. Type the following command:

ipconfig /renew

4. Check whether the laptop receives a valid IP address.
5. Make sure DHCP is enabled on the router.

4. Router or ISP Problem

Problem: The router may have lost its internet connection, or the Internet Service Provider (ISP) may be experiencing a service issue.

Solution:

1. Check whether other devices can access the internet.
2. Check the Internet/WAN indicator on the router.
3. Restart the broadband modem and Wi-Fi router.
4. Contact the ISP if the problem continues.

5. Wi-Fi Adapter or Driver Problem

Problem: The laptop's wireless network adapter may be disabled or its driver may not be working properly.

Solution:

1. Open Device Manager.
2. Expand Network Adapters.
3. Check whether the Wi-Fi adapter is enabled.
4. Update or reinstall the wireless driver if required.
5. Restart the laptop and test the connection.

6. Useful Troubleshooting Commands

The following commands can help diagnose network connectivity problems:

Command	Purpose
ipconfig	Displays IP configuration
ipconfig /release	Releases the current DHCP IP address
ipconfig /renew	Requests a new IP address
ping 192.168.1.1	Tests connectivity to the router
ping 8.8.8.8	Tests internet connectivity using an IP address
nslookup google.com	Checks DNS resolution

Conclusion

Network troubleshooting helps identify and resolve internet connectivity problems. By checking Wi-Fi settings, IP configuration, router connectivity, ISP services and network drivers, most common network issues can be diagnosed and fixed.